

GED Mathematics: Geometry Unit

November 6, 2025

Lesson 1: Points, Lines, and Angles

- Points represent exact locations in space.
- Lines extend infinitely in two directions.
- Line segments have two endpoints.
- Rays start at one point and extend infinitely in one direction.
- Angles are formed by two rays sharing a common endpoint.

Lesson 1: Worked Examples

Example 1: Find the complement of a 35° angle.

$$\text{Complement} = 90^\circ - 35^\circ = 55^\circ.$$

Example 2: Find the supplement of a 120° angle.

$$\text{Supplement} = 180^\circ - 120^\circ = 60^\circ.$$

Example 3: If two angles form a linear pair and one is 70° , the other is 110° .

Example 4: If two complementary angles are in the ratio 1:2, find them.

$$x + 2x = 90 \Rightarrow 3x = 90 \Rightarrow x = 30. \text{ So angles are } 30^\circ \text{ and } 60^\circ.$$

Example 5: Identify adjacent vs. vertical angles: Vertical angles are opposite each other when two lines intersect.

Lesson 2: Triangles

- The sum of angles in a triangle is 180° .
- Types: scalene, isosceles, equilateral.
- Pythagorean theorem: $a^2 + b^2 = c^2$.
- Area formula: $A = \frac{1}{2}bh$.
- Perimeter is the sum of side lengths.

Lesson 2: Worked Examples

Example 1: Find the missing angle in a triangle with angles 70° and 50° .

$$180 - (70 + 50) = 60^\circ.$$

Example 2: A right triangle has legs 3 and 4. Find the hypotenuse.

$$c = \sqrt{3^2 + 4^2} = 5.$$

Example 3: Find the area of a triangle with base 10 and height 6.

$$A = \frac{1}{2}(10)(6) = 30.$$

Example 4: For an isosceles triangle with equal sides 5 and base 6, find the height.

$$\text{Height} = \sqrt{5^2 - 3^2} = 4.$$

Example 5: Find the perimeter of a triangle with sides 5, 7, and 9.

$$5 + 7 + 9 = 21.$$

Lesson 3: Quadrilaterals

- Quadrilaterals have 4 sides and 360° in total.
- Types: square, rectangle, parallelogram, trapezoid, rhombus.
- Area formulas:
 - Rectangle: $A = lw$
 - Parallelogram: $A = bh$
 - Trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$

Lesson 3: Worked Examples

Example 1: Find the area of a rectangle with $l = 8$, $w = 3$.

$$A = 8 \times 3 = 24.$$

Example 2: Area of a parallelogram with base 12 and height 5:

$$A = 60.$$

Example 3: Area of a trapezoid with bases 10, 6 and height 4:

$$A = \frac{1}{2}(10 + 6)(4) = 32.$$

Example 4: A square has diagonal 10. Find side length:

$$s = \frac{10}{\sqrt{2}} = 5\sqrt{2}.$$

Example 5: Perimeter of rhombus with side 7: $4 \times 7 = 28$.

Lesson 4: Circles

- Radius: distance from center to any point on circle.
- Diameter: twice the radius.
- Circumference: $C = 2\pi r$.
- Area: $A = \pi r^2$.

Lesson 4: Worked Examples

Example 1: Find circumference with $r = 7$. $C = 2\pi(7) = 14\pi$.

Example 2: Find area with $r = 5$. $A = 25\pi$.

Example 3: If circumference = 31.4, find r . $r = \frac{31.4}{2\pi} = 5$.

Example 4: A circle has area 78.5. Find r . $r = \sqrt{\frac{78.5}{\pi}} \approx 5$.

Example 5: Diameter = 10. Find circumference: $C = \pi d = 10\pi$.

Lesson 5: 3D Figures

- Common solids: prism, cylinder, pyramid, cone, sphere.
- Volume formulas:
 - Prism: $V = Bh$
 - Cylinder: $V = \pi r^2 h$
 - Pyramid: $V = \frac{1}{3}Bh$
 - Cone: $V = \frac{1}{3}\pi r^2 h$
 - Sphere: $V = \frac{4}{3}\pi r^3$

Lesson 5: Worked Examples

Example 1: Find volume of a rectangular prism with $l = 4$, $w = 3$, $h = 5$.
 $V = lwh = 60$.

Example 2: Cylinder with $r = 3$, $h = 10$: $V = 90\pi$.

Example 3: Cone with $r = 4$, $h = 9$: $V = \frac{1}{3}\pi(16)(9) = 48\pi$.

Example 4: Sphere with $r = 6$: $V = \frac{4}{3}\pi(216) = 288\pi$.

Example 5: Pyramid with base area 30 and height 12: $V = 120$.